

**A**rtificial intelligence (AI) tools and technologies are used by some of the world's leading airliner service providers to deliver more personalised travelling experiences to their customers. AI, machine learning, machine vision, robotics and natural language processing are the future of the aviation industry. Predictive analytics, customer feedback analysis, auto-scheduling, pattern recognition and targeted advertising are all part of AI technology taking the aviation industry to a whole new level.

AI technology is becoming more advanced and complex, and is finding applications in the aviation industry in multiple ways, including auto-pilot features. Due to factors like cost sav-

tions and more on Internet-enabled devices, and so on.

AI can also be used in air cargo for such purposes as revenue and safety management.

**Check-in.** Checking in at the airport before boarding the flight is a vital task. AI technology can be used for this purpose in addition to identifying passengers. Many airlines are already using AI for online check-in via mobile apps and ticketing kiosks for a better flight experience.

AI technology will make the process of passenger identification faster and easier. Security scanners and biometric identification with AI will also help security personnel on duty to be more efficient.

aircraft maintenance. Use of AI makes the process of maintenance easy and reduces failure rates.

**Flight engineers and pilots.** A flight engineer is a member of an aircraft's flight crew, who monitors and operates complex aircraft systems. He or she is also responsible for pre- and post-flight aircraft inspections, and ensuring that weight and balance of aircraft is correctly calculated to ensure centre of gravity is within limits.

Now, the flight engineer's role has been replaced largely by computerised engine systems. Any malfunction, abnormality or emergency is displayed on an electronic display panel, and the computer automatically initiates corrective action to rectify the abnormal condition.

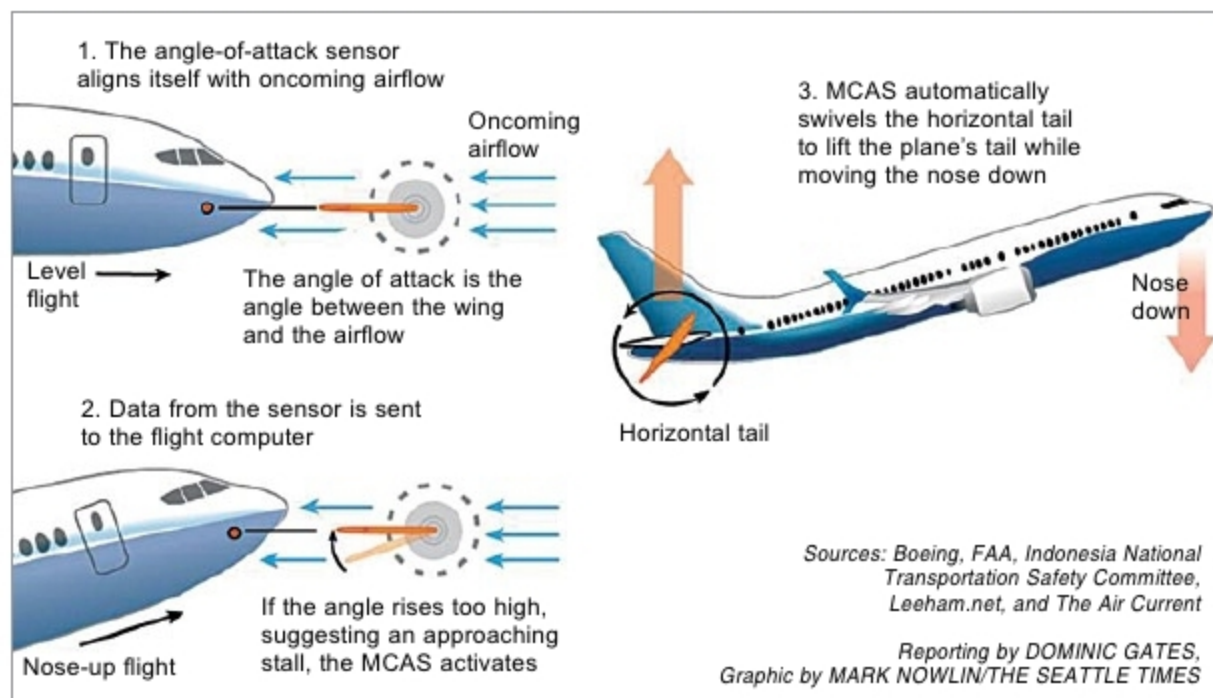
While some non-pilot crew members have been replaced by AI, human pilots still maintain their skills in the cockpit, monitoring the systems. **With advanced AI technology, hands-on piloting by human pilots might be reduced. Then, human pilots might be able to issue commands to the computer rather than having direct manual control.**

**Air traffic control.** One of the biggest challenges at the airport is changing weather and height of the control tower that can cause delays in air traffic. During poor weather, air traffic control (ATC) must rely on radars to keep the airport running smoothly.

In an effort to solve this problem, some airports have installed ultra-high-definition cameras with AI technology on top of the towers. AI can be used to deliver clear views of the airport to traffic controllers. AI platforms like AIMEE, developed by Searidge Technologies, utilise machine learning to interpret images, record aircraft and notify controllers so that they can signal the next plane for arrival on the cleared runway.

**Bomber pilots.** During the Gulf War, 297 missions were flown by AI-based autonomous aircraft without anyone onboard or any remote human pilot controlling them. Many explosive payloads were delivered to targets by such aircraft that flew hundreds of kilometres.

**Ground attack pilots.** Some ground attacks are now carried by unmanned combat aerial vehicles (UCAVs) managed by remote pilots instead of



How the maneuvering characteristics augmentation system (MCAS) works on Boeing 737 Max (Credit: [www.seattletimes.com](http://www.seattletimes.com))

ings and shortage of qualified pilots, many companies have expressed an interest in reducing or even eliminating the number of pilots in the cockpit. There have been speculations that AI may someday replace human pilots. **This article covers the various applications of AI in aviation and discusses how close AI is to replacing human pilots in the future.**

### AI in aviation industry

Some of the areas where AI is being, and can be, applied in the aviation sector are:

**Customer service agents.** AI can be used to assist customers in many ways, including assisting check-in requests, providing accurate information on future flights and status of current flights, answering common ques-

**Baggage screening.** Baggage screening is another important task at any airport. AI can help duty managers simplify this process. AI-based technology like Syntech ONE comes handy in such applications. It can detect threats at x-ray and CT security checkpoints with automated screening. The system can be installed at airports to screen baggage in multiple passenger lanes.

**Maintenance prediction.** AI technology can also be used to predict potential failure of aircraft. Leading aircraft manufacturers use cloud-based data storing systems to improve reliability of aircraft maintenance. This allows the fleet to collect and record a huge amount of real-time data. AI-based predictive analysis provides a systematic approach for